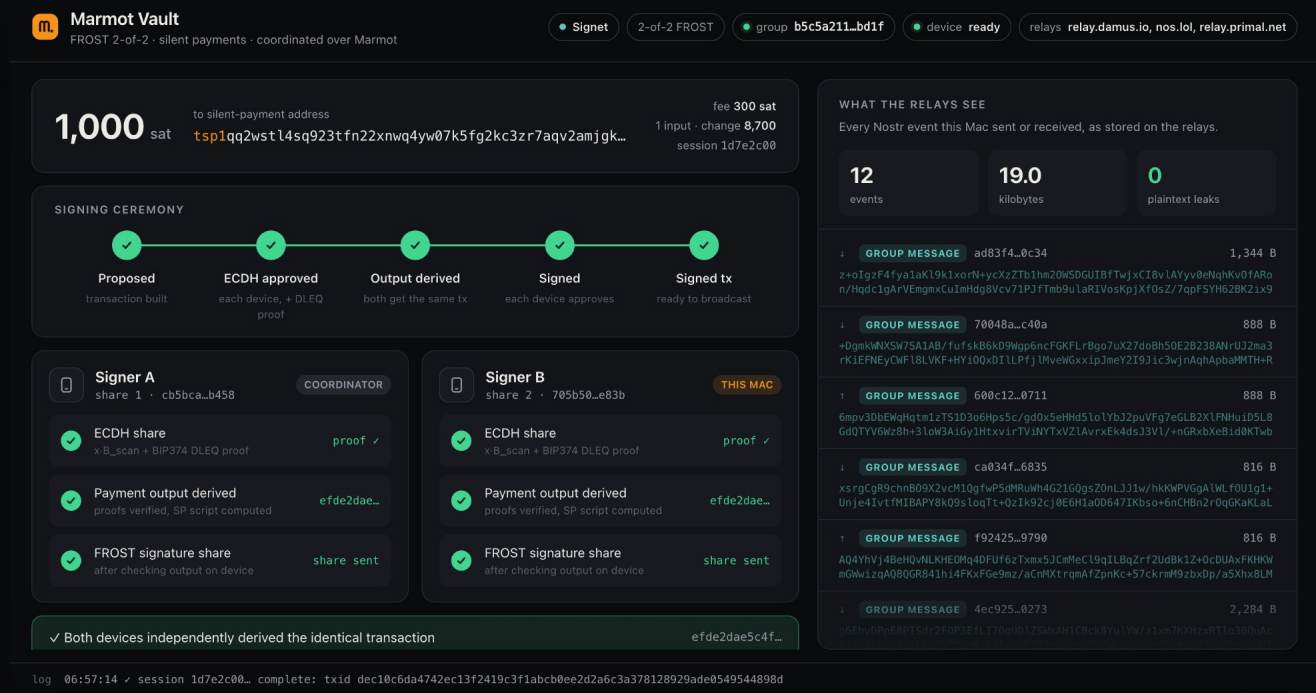
BITCOIN++ SEOUL · BEST USE OF MARMOT

Black·Ice

Distributed Frost Silent Payments

Hardware signers on separate devices co-sign a bitcoin payment to a Silent Payment address, coordinated over an encrypted Marmot group.



BlackIce: Robos&Rademenez · Signet · November 2026

Silent payments, now from a hardware multisig

UNTIL NOW

Single-key wallets only

To pay a silent-payment address, the sender needs the combined secret of every input key. A multisig never has that in one place, so no multisig could send to one.

WITH BLACKICE

Hardware multisig pays them

Each Frostsnap device contributes its own ECDH share with a proof. The shares combine into the payment output, and no key ever leaves its device.

Frostsnap hardware

FROST threshold keys, any k-of-n

Confirmed on Signet

Three layers, each one private

MARMOT

Encrypted coordination

MLS group messaging over Nostr. Relays store only ciphertext. No phone number, no server.

Fixes the chat

FROST 2-OF-2

One key, one signature

Two devices sign as one Schnorr key. On-chain it’s an ordinary single-key taproot spend.

Fixes the chain

SILENT PAYMENTS

Unlinkable outputs

One static address. Each payment lands in a fresh output that nobody else can link to it.

Fixes the recipient

Until now, only single-key wallets could pay a silent-payment address. **BlackIce sends from a hardware multisig.**

How a payment gets signed

1

Propose

Signer A's bridge builds the transaction. The payment output can't exist yet.

2

ECDH + proof

Each device asks its owner, then returns its ECDH share with a BIP374 DLEQ proof.

3

Derive

Each bridge verifies both proofs and computes the payment output. Both results must match.

4

Sign

Each device shows the real output. Its owner approves and it returns a FROST share.

5

Combine

Two shares become one Schnorr signature, ready to broadcast.

Every message between steps is an MLS-encrypted Marmot message on public Nostr relays.

STEP 1 / 7

In the encrypted Marmot group, waiting for a proposal

Marmot Vault

FROST 2-of-2 · silent payments · coordinated over Marmot

Signet | 2-of-2 FROST | group b5c5a211…bd1f | device ready | relays relay.damus.io, nos.lol, relay.primal.net

Waiting for a proposal from the coordinator...

SIGNING CEREMONY

1. **Proposed** — transaction built
2. **ECDH approved** — each device, + DLEQ proof
3. **Output derived** — both get the same tx
4. **Signed** — each device approves
5. **Signed tx** — ready to broadcast

Signer A

share 1 · cb5bca…b458 — COORDINATOR

- ECDH share — x·B_scan + BIP374 DLEQ proof
- Payment output derived — proofs verified, SP script computed
- FROST signature share — after checking output on device

Signer B

share 2 · 705b50…e83b — THIS MAC

- ECDH share — x·B_scan + BIP374 DLEQ proof
- Payment output derived — proofs verified, SP script computed
- FROST signature share — after checking output on device

WHAT THE RELAYS SEE

Every Nostr event this Mac sent or received, as stored on the relays.

3	6.9	0
events	kilobytes	plaintext leaks

↓ GIFT-WRAPPED INVITE 5bf02b…fa22 5,552 B

```
ArXDqWAwntqxU4Nqs3QFG5zyOTYK2D5R3cpjuaxvh4JvAsuD9vT5I/yW010yDSepPWMm
VHs+Cn3Zdr+mG8w0kVmxsbn+7nvEI9BU6+bCpJYSjLnHswjrjNmDdN9RjudXPBFMGhjh
```

↓ GROUP MESSAGE 56c89c…eadb 924 B

```
BIfx+WB6xwLHUfBWHQ8RTqORaGM+QRJnNAfWrtUiQW0ac8sLw9sXOPSt+Vx5l3V9hwWf
YFQg/C/iU9r9cCTDj6CmsfB+2B/PYd9HlkOtFrvCogqE43dzxaMc4OuaT/Hoz8hK7Knc
```

↑ KEY PACKAGE 705b50…e83b 632 B

```
AAEABQABAAEgpH1hioUcS6KYrcdoPIMnp3hgCtvJqVzDk17WNv3chyogBoB6VOJkW/Bn
gvfTkGRS4YVhbhyLzRoACTRyJVLp9w0gFyyBmJrKIfMeSa88W0bRylGiDamPWQPDAy5i
```

log 06:51:54 ✓ device 02c622f7… (Signer B) connected with the expected SP firmware

STEP 2 / 7

1,000 sat to a silent-payment address: both devices ask to approve

Marmot Vault

FROST 2-of-2 · silent payments · coordinated over Marmot

Signet | 2-of-2 FROST | group b5c5a211…bd1f | device ready | relays relay.damus.io, nos.lol, relay.primal.net

1,000 sat

to silent-payment address

`tsp1qq2wstl4sq923tfn22xnwq4yw07k5fg2kc3zr7aqv2amjgk…`

fee **300 sat**
1 input · change **8,700**
session `1d7e2c00`

SIGNING CEREMONY

1. ✓ **Proposed** — transaction built
2. ~~2~~ **ECDH approved** — each device, + DLEQ proof
3. 3 **Output derived** — both get the same tx
4. 4 **Signed** — each device approves
5. 5 **Signed tx** — ready to broadcast

Signer A

share 1 · cb5bca…b458 — COORDINATOR

- ECDH share — x·B_scan + BIP374 DLEQ proof — Approve on device
- Payment output derived — proofs verified, SP script computed
- FROST signature share — after checking output on device

Signer B

share 2 · 705b50…e83b — THIS MAC

- ECDH share — x·B_scan + BIP374 DLEQ proof — Approve on device
- Payment output derived — proofs verified, SP script computed
- FROST signature share — after checking output on device

WHAT THE RELAYS SEE

Every Nostr event this Mac sent or received, as stored on the relays.

4	9.3	0
events	kilobytes	plaintext leaks

↓ GROUP MESSAGE `04273c…88c1` 2,460 B

```
4hlQWw3dqbUphh70QJDWRMgSWyRI259u3VOr97VUATDuJjGshmzYSn262xJDIQC2TNqf
D/nX4hT/zYzsKzHp2XDN7MEQ3MWn9f5dkzbNtmpZ3fc2HuEJZ0FyxCEwy6lUAp39jacn
```

↓ GIFT-WRAPPED INVITE `5bf02b…fa22` 5,552 B

```
ArXDqWAwntqxU4Nqs3QFG5zyOTYK2D5R3cpjuaxvh4JvAsuD9vT5I/yW010yDSepPWMm
VHs+Cn3Zdr+mG8w0kVmxsbn+7nvEI9BU6+bCpJYSjLnHswjrjNmDdN9RjudXPBFMGhjh
```

↓ GROUP MESSAGE `56c89c…eadb` 924 B

```
BIfx+WB6xwLHUfBWHQ8RTqORaGM+QRJnNAfWrtUiQW0ac8sLw9sXOPSt+Vx5l3V9hwWf
YFQg/C/iU9r9cCTDj6CmsfB+2B/PYd9HlkOtFrvCogqE43dzxaMc4OuaT/Hoz8hK7Knc
```

↑ KEY PACKAGE `705b50…e83b` 632 B

```
AAEABQABAAEgpH1hioUcS6KYrcdoPIMnp3hgCtvJqVzDk17WNv3chyogBoB6VOJkW/Bn
gvfTkGRS4YVhbhyLzRoACTRyJVLp9w0gFyyBmJrKIfMeSa88W0bRylGiDamPWQPDAy5i
```

log 06:54:04 ecdh: confirm_ecdh_on_device

STEP 3 / 7

A approved ECDH; B checks the recipient on its device

Marmot Vault

FROST 2-of-2 · silent payments · coordinated over Marmot

Signet | 2-of-2 FROST | group b5c5a211…bd1f | device ready | relays relay.damus.io, nos.lol, relay.primal.net

1,000 sat

to silent-payment address

tsp1qq2wstl4sq923tfn22xnwq4yw07k5fg2kc3zr7aqv2amjgk…

fee **300** sat
1 input · change **8,700**
session 1d7e2c00

SIGNING CEREMONY

✓	2	3	4	5
Proposed	ECDH approved	Output derived	Signed	Signed tx
transaction built	each device, + DLEQ proof	both get the same tx	each device approves	ready to broadcast

Signer A

share 1 · cb5bca…b458 — COORDINATOR

- ECDH share — x·B_scan + BIP374 DLEQ proof — proof ✓
- Payment output derived — proofs verified, SP script computed
- FROST signature share — after checking output on device

Signer B

share 2 · 705b50…e83b — THIS MAC

- ECDH share — x·B_scan + BIP374 DLEQ proof — Approve on device
- Payment output derived — proofs verified, SP script computed
- FROST signature share — after checking output on device

WHAT THE RELAYS SEE

Every Nostr event this Mac sent or received, as stored on the relays.

5	10.8	0
events	kilobytes	plaintext leaks

↓ GROUP MESSAGE c1365a…75f5 — 1,448 B

```
EY4qGIaA669VenoocTF+yLKyJEua6taViAZ1aC8LEPRzKrKOMuHLz9ctTVZ0kKkV5p31
P5K0Q8j2AX9wkE5gWVoX6nE3n6+23xSUH6/Yng4K0byx6u2OcHc96ZDcMxC2OC04G5ZB
```

↓ GROUP MESSAGE 04273c…88c1 — 2,460 B

```
4hlQWw3dqbUphh70QJDWRMgSWyRI259u3VOr97VUATDuJjGshmzYSn262xJDIQC2TNqf
D/nX4hT/zYzsKzHp2XDN7MEQ3MWn9f5dkzbNtmpZ3fc2HuEJZ0FyxCEwy6lUAp39jacn
```

↓ GIFT-WRAPPED INVITE 5bf02b…fa22 — 5,552 B

```
ArXDqWAwntqxU4Nqs3QFG5zyOTYK2D5R3cpjuaxvh4JvAsuD9vT5I/yW010yDSepPWMm
VHs+Cn3Zdr+mG8w0kVmxsbn+7nvEI9BU6+bCpJYSjLnHswjrjNmDdN9RjudXPBFMGhjh
```

↓ GROUP MESSAGE 56c89c…eadb — 924 B

```
BIfx+WB6xwLHUfBWHQ8RTqORaGM+QRJnNAfWrtUiQW0ac8sLw9sXOPSt+Vx5l3V9hwWf
YFQg/C/iU9r9cCTDj6CmsfB+2B/PYd9HlkOtFrvCogqE43dzxaMc4OuaT/Hoz8hK7Knc
```

↑ KEY PACKAGE 705b50…e83b — 632 B

```
AAEABQABAAEgpH1hioUcS6KYrcdoPIMnp3hgCtvJqVzDk17WNv3chyogBoB6VOJkW/Bn
gvfTkGRS4YVhbhyLzRoACTRyJVLp9w0gFyyBmJrKIfMeSa88W0bRylGiDamPWQPDAy5i
```

log 06:54:04 ecdh: confirm_ecdh_on_device

STEP 4 / 7

Both ECDH shares and DLEQ proofs exchanged over Marmot

Marmot Vault

FROST 2-of-2 · silent payments · coordinated over Marmot

Signet | 2-of-2 FROST | group b5c5a211…bd1f | device ready | relays relay.damus.io, nos.lol, relay.primal.net

1,000 sat

to silent-payment address

tsp1qq2wstl4sq923tfn22xnwq4yw07k5fg2kc3zr7aqv2amjgk…

fee 300 sat
1 input · change 8,700
session 1d7e2c00

SIGNING CEREMONY

1	2	3	4	5
Proposed	ECDH approved	Output derived	Signed	Signed tx
transaction built	each device, + DLEQ proof	both get the same tx	each device approves	ready to broadcast

Signer A

share 1 · cb5bca…b458 — COORDINATOR

- ECDH share — x·B_scan + BIP374 DLEQ proof — proof ✓
- Payment output derived — proofs verified, SP script computed
- FROST signature share — after checking output on device

Signer B

share 2 · 705b50…e83b — THIS MAC

- ECDH share — x·B_scan + BIP374 DLEQ proof — proof ✓
- Payment output derived — proofs verified, SP script computed
- FROST signature share — after checking output on device

✓ Both devices approved ECDH and sent their DLEQ proofs — waiting for the coordinator to request signatures

WHAT THE RELAYS SEE

Every Nostr event this Mac sent or received, as stored on the relays.

6	12.2	0
events	kilobytes	plaintext leaks

↑ GROUP MESSAGE a3e5a2…108e 1,448 B

3BPW20EW4H7c2zks6PycIjeaggGT9OMldWioF45heeav3liefUaJDDeUWgH3QjSDD9h+
8cQVQ9T2Z/QtXxh8B70PnWxLNBk0FpMX2DZ0UCX9tg0vwb2EbANM3ra4pE3SGj2x3R90

↓ GROUP MESSAGE c1365a…75f5 1,448 B

EY4qGIaA669VenoocTF+yLKyJEua6taViAZ1aC8LEPRzKrKOMuHLz9ctTVZ0kKkV5p31
P5K0Q8j2AX9wkE5gWVoX6nE3n6+23xSUH6/Yng4K0byx6u2OcHc96ZDcMxC2OC04G5ZB

↓ GROUP MESSAGE 04273c…88c1 2,460 B

4hlQWw3dqbUphh70QJDWRMgSWyRI259u3VOr97VUATDuJjGshmzYSn262xJDIQC2TNqf
D/nX4hT/zYzsKzHp2XDN7MEQ3MWn9f5dkzbNtmpZ3fc2HuEJZ0FyxCEwy6lUAp39jacn

↓ GIFT-WRAPPED INVITE 5bf02b…fa22 5,552 B

ArXDqWAwntqxU4Nqs3QFG5zyOTYK2D5R3cpjuaxvh4JvAsuD9vT5I/yW010yDSepPWMm
VHs+Cn3Zdr+mG8w0kVmxsbn+7nvEI9BU6+bCpJYSjLnHswjrjNmDdN9RjudXPBFMGhjh

↓ GROUP MESSAGE 56c89c…eadb 924 B

BIfx+WB6xwLHUfBWHQ8RTqORaGM+QRJnNAfWrtUiQW0ac8sLw9sXOPSt+Vx5l3V9hwWf
YFQg/C/iU9r9cCTDj6CmsfB+2B/PYd9HlkOtFrvCogqE43dzxaMc4OuaT/Hoz8hK7Knc

↑ KEY PACKAGE 705b50…e83b 632 B

AAEABQABAAEgpH1hioUcS6KYrcdoPIMnp3hgCtvJqVzDk17WNv3chyogBoB6VOJkW/Bn

log 06:55:27 ✓ device approved ECDH — sending offer (share 2)

STEP 5 / 7

Both derived the identical tx; each device shows the output

Marmot Vault

FROST 2-of-2 · silent payments · coordinated over Marmot

Signet | 2-of-2 FROST | group b5c5a211…bd1f | device ready | relays relay.damus.io, nos.lol, relay.primal.net

1,000 sat

to silent-payment address

tsp1qq2wstl4sq923tfn22xnwq4yw07k5fg2kc3zr7aqv2amjgk…

fee 300 sat
1 input · change 8,700
session 1d7e2c00

SIGNING CEREMONY

1	2	3	4	5
Proposed	ECDH approved	Output derived	Signed	Signed tx
transaction built	each device, + DLEQ proof	both get the same tx	each device approves	ready to broadcast

Signer A

share 1 · cb5bca…b458 — COORDINATOR

- ECDH share — x·B_scan + BIP374 DLEQ proof — proof ✓
- Payment output derived — proofs verified, SP script computed — efde2dae…
- FROST signature share — after checking output on device — Approve on device

Signer B

share 2 · 705b50…e83b — THIS MAC

- ECDH share — x·B_scan + BIP374 DLEQ proof — proof ✓
- Payment output derived — proofs verified, SP script computed — efde2dae…
- FROST signature share — after checking output on device — Approve on device

✓ Both devices independently derived the identical transaction — efde2dae5c4f…

WHAT THE RELAYS SEE

Every Nostr event this Mac sent or received, as stored on the relays.

9	16.0	0
events	kilobytes	plaintext leaks

↓	GROUP MESSAGE	ca034f…6835	816 B

xsrgCgR9chnBO9X2vcM1QgfwP5dMRuWh4G21GQgsZOnLJJ1w/hkKWPVGgAlWLfOU1g1+Unje4IvtfMIBAPY8kQ9sloqTt+QzIk92cj0E6H1aOD647IKbso+6nCHBn2rOqGKaKLaL

↑	GROUP MESSAGE	f92425…9790	816 B

AQ4YhVj4BeHQvNLKHEOMq4DFUf6zTxmx5JCmMeCl9qILBqZrf2UdBk1Z+OcDUAxFKHKWmGWwizqAQ8QGR841hi4FKxFGe9mz/aCnMXtrqmAfZpnKc+57ckrmM9zbxDp/a5Xhx8LM

↓	GROUP MESSAGE	4ec925…0273	2,284 B

g6EhvDPnE8PTSdr2FOP3EfLI7OgUDlZSWxAH1CBck8YulYW/z1xm7KXHzxRTlo36QuAcAIoaashgh1y1GbaRNTMVcB/bfI+zYP3S3WxgMKa9kYwNS8Ujre0D+mogD2oVKDp6UcNl

↑	GROUP MESSAGE	a3e5a2…108e	1,448 B

3BPW20EW4H7c2zks6PycIjeaggGT9OMldWioF45heeav3liefUaJDDeUWgH3QjSDD9h+8cQVQ9T2Z/QtXxh8B70PnWxLNBk0FpMX2DZ0UCX9tg0vwb2EbANM3ra4pE3SGj2x3R90

↓	GROUP MESSAGE	c1365a…75f5	1,448 B

EY4qGIaA669VenoocTF+yLKyJEua6taViAZ1aC8LEPRzKrKOMuHLz9ctTVZ0kKkV5p31P5K0Q8j2AX9wkE5gWVoX6nE3n6+23xSUH6/Yng4K0byx6u2OcHc96ZDcMxC2OC04G5ZB

↓	GROUP MESSAGE	04273c…88c1	2,460 B

4hlQWw3dqbUphh70QJDWRMgSWyRI259u3VOr97VUATDuJjGshmzYSn262xJDIQC2TNqf

log 06:55:56 sign: confirm_transaction_on_device

STEP 6 / 7

B signed; A approves on its device

Marmot Vault

FROST 2-of-2 · silent payments · coordinated over Marmot

Signet | 2-of-2 FROST | group b5c5a211…bd1f | device ready | relays relay.damus.io, nos.lol, relay.primal.net

1,000 sat

to silent-payment address

tsp1qq2wstl4sq923tfn22xnwq4yw07k5fg2kc3zr7aqv2amjgk…

fee **300** sat
1 input · change **8,700**
session 1d7e2c00

SIGNING CEREMONY

1. **Proposed** — transaction built
2. **ECDH approved** — each device, + DLEQ proof
3. **Output derived** — both get the same tx
4. **Signed** — each device approves
5. **Signed tx** — ready to broadcast

Signer A

share 1 · cb5bca…b458 — COORDINATOR

- ECDH share — x·B_scan + BIP374 DLEQ proof — proof ✓
- Payment output derived — proofs verified, SP script computed — efde2dae…
- FROST signature share — after checking output on device — Approve on device

Signer B

share 2 · 705b50…e83b — THIS MAC

- ECDH share — x·B_scan + BIP374 DLEQ proof — proof ✓
- Payment output derived — proofs verified, SP script computed — efde2dae…
- FROST signature share — after checking output on device — share sent

✓ Both devices independently derived the identical transaction efde2dae5c4f…

WHAT THE RELAYS SEE

Every Nostr event this Mac sent or received, as stored on the relays.

10	16.9	0
events	kilobytes	plaintext leaks

↑ GROUP MESSAGE 600c12…0711 888 B
6mpv3DbEWqHqtm1zTS1D3o6Hps5c/gdOx5eHHd5lolYbJ2puVFg7eGLB2XlFNHuiD5L8
GdQTYV6Wz8h+3loW3AiGy1HtxvirTViNYTxVZlAvrxEk4dsJ3Vl/+nGRxbXeBid0KTwb

↓ GROUP MESSAGE ca034f…6835 816 B
xsrgCgR9chnBO9X2vcM1QgfwP5dMRuWh4G21GQgsZOnLJJ1w/hkKWPVGgAlWLfOU1g1+
Unje4IvtfMIBAPY8kQ9sloqTt+QzIk92cj0E6H1aOD647IKbso+6nCHBn2rOqGKaKLaL

↑ GROUP MESSAGE f92425…9790 816 B
AQ4YhVj4BeHQvNLKHEOMq4DFUf6zTxmx5JCmMeCl9qILBqZrf2UdBk1Z+OcDUAxFKHKW
mGWwizqAQ8QGR841hi4FKxFGe9mz/aCnMXtrqmAfZpnKc+57ckrmM9zbxDp/a5Xhx8LM

↓ GROUP MESSAGE 4ec925…0273 2,284 B
g6EhvDPnE8PTSdr2FOP3EfLI7OgUDlZSWxAH1CBck8YulYW/z1xm7KXHzxRTlo36QuAc
AIoaashgh1y1GbaRNTMVcB/bfI+zYP3S3WxgMKa9kYwNS8Ujre0D+mogD2oVKDp6UcNl

↑ GROUP MESSAGE a3e5a2…108e 1,448 B
3BPW20EW4H7c2zks6PycIjeaggGT9OMldWioF45heeav3liefUaJDDeUWgH3QjSDD9h+
8cQVQ9T2Z/QtXxh8B70PnWxLNBk0FpMX2DZ0UCX9tg0vwb2EbANM3ra4pE3SGj2x3R90

↓ GROUP MESSAGE c1365a…75f5 1,448 B
EY4qGIaA669VenoocTF+yLKyJEua6taViAZ1aC8LEPRzKrKOMuHLz9ctTVZ0kKkV5p31

log 06:56:54 ✓ device signed — sending shares (share 2)

STEP 7 / 7

One Schnorr signature. 0 plaintext leaks on the relays

Marmot Vault

FROST 2-of-2 · silent payments · coordinated over Marmot

Signet | 2-of-2 FROST | group b5c5a211…bd1f | device ready | relays relay.damus.io, nos.lol, relay.primal.net

1,000 sat

to silent-payment address

tsp1qq2wstl4sq923tfn22xnwq4yw07k5fg2kc3zr7aqv2amjgk…

fee 300 sat
1 input · change 8,700
session 1d7e2c00

SIGNING CEREMONY

Proposed	ECDH approved	Output derived	Signed	Signed tx
transaction built	each device, + DLEQ proof	both get the same tx	each device approves	ready to broadcast

Signer A

share 1 · cb5bca…b458 — COORDINATOR

ECDH share	x·B_scan + BIP374 DLEQ proof	proof ✓
Payment output derived	proofs verified, SP script computed	efde2dae…
FROST signature share	after checking output on device	share sent

Signer B

share 2 · 705b50…e83b — THIS MAC

ECDH share	x·B_scan + BIP374 DLEQ proof	proof ✓
Payment output derived	proofs verified, SP script computed	efde2dae…
FROST signature share	after checking output on device	share sent

✓ Both devices independently derived the identical transaction efde2dae5c4f…

WHAT THE RELAYS SEE

Every Nostr event this Mac sent or received, as stored on the relays.

12	19.0	0
events	kilobytes	plaintext leaks

↓ GROUP MESSAGE ad83f4…0c34 1,344 B

z+oIgzF4fya1aKl9k1xorN+ycXzZTb1hm2OWSDGUIBfTwjxCI8vlAYyv0eNqhKvOfARo
n/Hqdc1gArVEmgmxCuImHdg8Vcv71PJfTmb9ulaRIVosKpjXfOsZ/7qpFSYH62BK2ix9

↓ GROUP MESSAGE 70048a…c40a 888 B

+DgmkWNXSW7SA1AB/fufskB6kD9Wgp6ncFGKFLrBgo7uX27doBh5OE2B238ANrUJ2ma3
rKiEFNEyCWFl8LVKF+HYiOQxDIlLPfjlMveWGxxipJmeY2I9Jic3wjnAqhApbaMMTH+R

↑ GROUP MESSAGE 600c12…0711 888 B

6mpv3DbEWqHqtm1zTS1D3o6Hps5c/gdOx5eHHd5lolYbJ2puVFg7eGLB2XlFNHuiD5L8
GdQTYV6Wz8h+3loW3AiGy1HtxvirTViNYTxVZlAvrxEk4dsJ3Vl/+nGRxbXeBid0KTwb

↓ GROUP MESSAGE ca034f…6835 816 B

xsrgCgR9chnBO9X2vcM1QgfwP5dMRuWh4G21GQgsZOnLJJ1w/hkKWPVGgAlWLfOU1g1+
Unje4IvtfMIBAPY8kQ9sloqTt+QzIk92cj0E6H1aOD647IKbso+6nCHBn2rOqGKaKLaL

↑ GROUP MESSAGE f92425…9790 816 B

AQ4YhVj4BeHQvNLKHEOMq4DFUf6zTxmx5JCmMeCl9qILBqZrf2UdBk1Z+OcDUAxFKHKW
mGWwizqAQ8QGR841hi4FKxFGe9mz/aCnMXtrqmAfZpnKc+57ckrmM9zbxDp/a5Xhx8LM

↓ GROUP MESSAGE 4ec925…0273 2,284 B

g6EhvDPnE8PTSdr2F0P3EfLI7OgUDlZSWxAH1CBck8YulYW/z1xm7KXHzxRTlo36QuAc

log 06:57:14 ✓ session 1d7e2c00… complete: txid dec10c6da4742ec13f2419c3f1abcb0ee2d2a6c3a378128929ade0549544898d

The device checks it, not the laptop



STEP 2 · ECDH

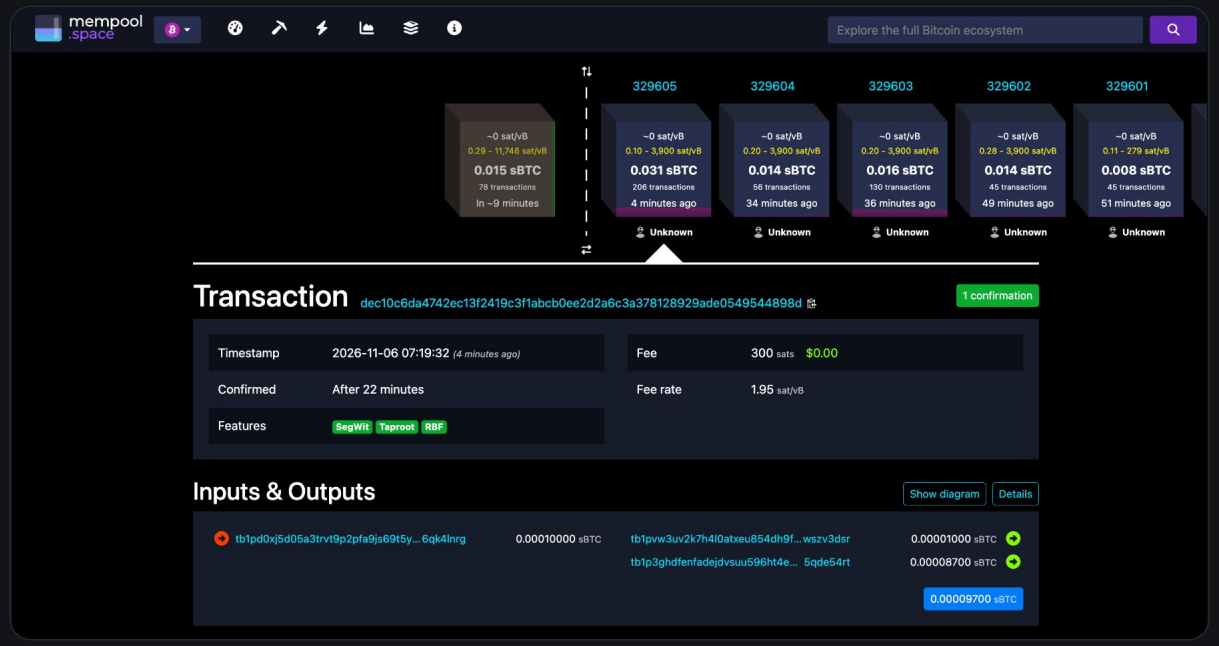
“Hold to Contribute” adds the device’s key share. It isn’t signing yet.



STEP 4 · SIGN

The device shows the silent-payment address, amount and fee, then “Hold to Sign”.

On-chain, it’s just a taproot spend



CONFIRMED · BLOCK 329,605

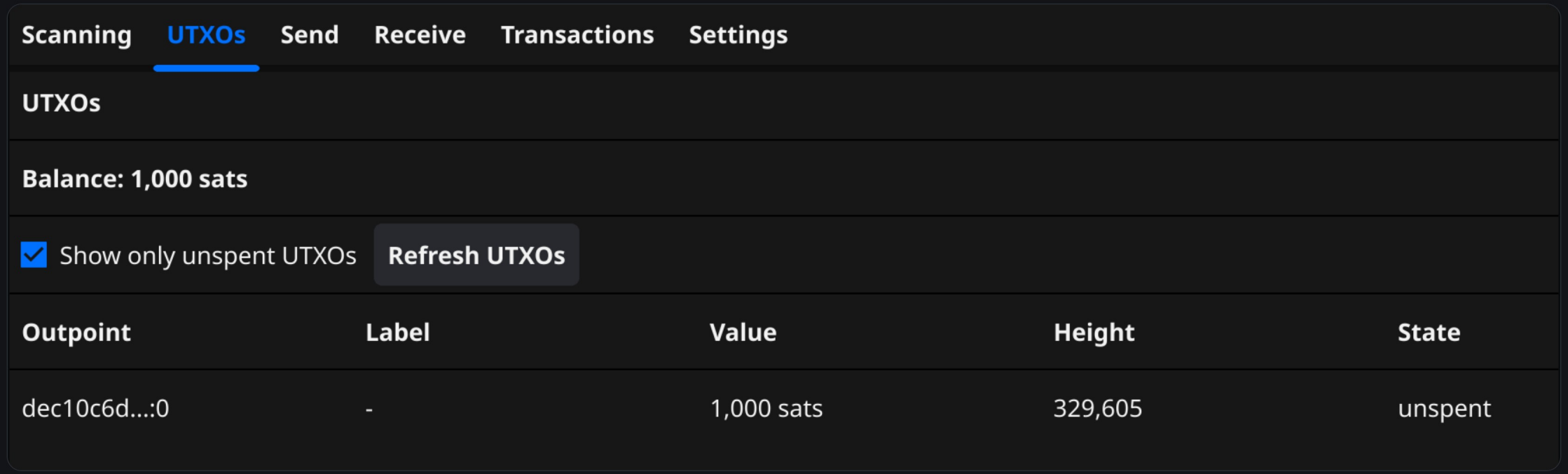
`dec10c6da4742ec13f2419c3`
`f1abcb0ee2d2a6c3a3781289`
`29ade0549544898d`

≈154 vB: one key-path input and two taproot outputs. No script, no second key, no multisig fingerprint.

The 1,000 sat landed at tb1pvw3u…v3dsr, a fresh taproot address nobody can link to the recipient’s tsp1… address.

mempool.space/signet · confirmed 2026-11-06 07:19 · fee 300 sat at 1.95 sat/vB

And the recipient’s wallet found it



Scanning Signet with its silent-payment keys, the recipient’s wallet picked up **1,000 sat** at **dec10c6d…:0**, block 329,605, ready to spend. Nobody else can tell that output belongs to them.

Recipient’s silent-payment wallet · UTXOs · same transaction as the previous slide

What’s new, and what’s next

BUILT THIS WEEKEND

- Remote FROST signing on real hardware, coordinated only over Marmot
- A hardware multisig that pays silent-payment addresses, with per-device DLEQ proofs (Frostsnap PR #640)
- No coordinator server; joining the group grants no signing power
- No signature share leaves a device until both derive the identical transaction

NEXT

- Import transactions from Sparrow
- k-of-n vaults, such as 2-of-3
- Receive to the vault’s own silent-payment address
- Fix the device’s intermittent serial crash upstream



Black·Ice

Distributed. Frost. Silent Payments.

No phone number. No server. No on-chain fingerprint.

`github.com/rademenezbtc/blackice`